# Platinum-loaded TiO2 for photocatalytic hydrogen evolution

We report a simple impregnation route to Pt-loaded TiO2 nanoparticles. The material shows a band gap of 3.2 eV and a hydrogen evolution rate of 450 umol/g/h under UV light. Stability was tested over five cycles of 4 h each without loss in activity.

| Sample | Band gap | H2 yield |
|---|---|---|
| TiO2 | 3.3 eV | 12 umol/g/h |
| TiO2-Pt | 3.2 eV | 450 umol/g/h |

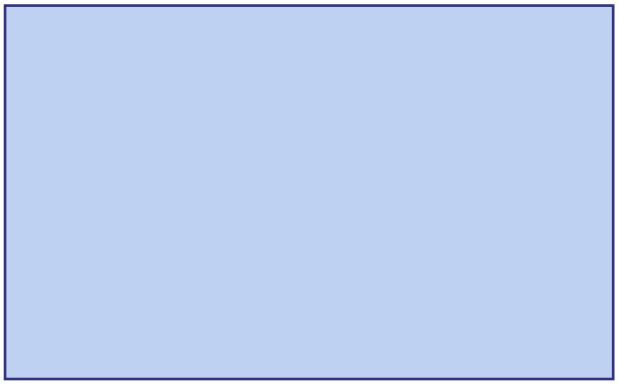

Figure 1: H2 evolution over time.

Conclusion: Pt loading increases activity by a factor of 37.